

GUILHERME AMARAL

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Summary

Systems engineer building low-level software and observability tooling for safety-critical RTOS, virtualization, and ARM/RISC-V SoCs. Experienced in secure-execution with ARM TrustZone, memory-profiling, and hardware/software debugging workflows.

Education

The University of Texas at Austin

B.S. in Computer Science

Aug. 2022 – May 2026

Austin, TX

Technical Skills

Languages / Systems: C/C++, Python, Java, AArch64, RISC-V, x86-64, QEMU, Unix/Linux

Technologies / Tools: Bash, Make, Valgrind, Windows, Jira, Splunk, Git, GitHub, Gerrit, Coverity, Jenkins

Experience

NVIDIA Corporation

Incoming Systems Software Engineer – RTOS and Virtualization

July 2026

Santa Clara, CA

NVIDIA Corporation

Systems Software Engineer Intern – RTOS and Virtualization

May 2025 – Aug. 2025

Santa Clara, CA

- Expanded production RTOS/virtualization memory-profiling infrastructure across NVIDIA DRIVE Thor/Orin SoCs and ARM/RISC-V clusters, tripling supported configurations and improving visibility into memory utilization across hardware/software boundaries.
- Delivered safety, security, and performance-focused proof-of-concepts with cross-functional teams, influencing product safety compliance decisions for RTOS/virtualization deployments.
- Designed high-reliability C++14 MMU & SMMU data-capture APIs across ARM TrustZone boundaries using FuSa libraries, enabling secure low-level observability for performance, safety, and debugging workflows.

NVIDIA Corporation

Systems Software Engineer Intern – RTOS and Virtualization

May 2024 – Aug. 2024

Santa Clara, CA

- Engineered hypervisor memory profiling tools to visualize RAM utilization across TrustZone layers, enabling developers to better use available memory on automotive platforms (2x improvement).
- Built ARMv8 GICv3 interrupt-profiling tooling to analyze SGI/PPI/SPI behavior across secure and non-secure execution contexts, helping SoC teams isolate hardware/software defects and performance bottlenecks.
- Shipped production C++14 to ISO 26262, AUTOSAR, MISRA, CERT compliance, validated through Jenkins/Splunk CI pipelines.

Technical Projects

Ext2 Filesystem | *x86, C++*

- Developed a full Ext2 filesystem in C++ for emulated x86 hardware on QEMU, supporting read/write operations for directories, files, and symbolic links up to 256 MiB.
- Designed a virtual memory subsystem with paging, fault handling, and file caching for kernel/user-space programs, reducing average memory access time by 80x.

QuickJS Runtime Exploit | *JavaScript, C*

- Exploited a use-after-free vulnerability in QuickJS to trigger type confusion and construct GC-resistant primitives: `addrof()`, `fakeobj()`, and `fakestr()`.
- Built `readmem()` and `writemem()` primitives to inspect PLT/GOT for reliable ASLR bypass, then marked pages RWX via `mprotect()` for second-stage payload execution.

Shellcode Encoder | *RISC-V, C*

- Engineered a UTF-8-compliant, non-NULL two-stage attack payload in assembly using a custom XOR-based shellcode encoder for 64-bit RISC-V targets.
- Optimized payload encodings to satisfy exploit constraints while reducing final binary size by 10% via bit-level transformations.

Interests

AI Infrastructure, AI Safety, Embedded Systems, Computer Architecture, Operating Systems, High-Performance Computing, Computer Security, Formal Methods